

Black tea increases coronary flow velocity reserve in healthy male subjects.



Epidemiologic studies suggest that tea consumption decreases the risk for cardiovascular events. However, there has been no clinical report examining the effects of tea consumption on coronary circulation. The purpose of this study was to evaluate the effects of black tea on coronary flow velocity reserve (CFVR) using transthoracic Doppler echocardiography (TTDE). This was a double-blind crossover study of 10 healthy male volunteers conducted to compare the effects of black tea and caffeine on coronary circulation. The coronary flow velocity of the left anterior descending coronary artery was measured at baseline and at hyperemia during adenosine triphosphate infusion by TTDE to determine CFVR. The CFVR ratio was defined as the ratio of CFVR after beverage consumption to CFVR before beverage consumption. All data were divided into 2 groups according to beverage type: group T (black tea) and group C (caffeine). Two-way analysis of variance showed a significant group effect and interaction in CFVR before and after beverage consumption ($p = 0.001$). CFVR significantly increased after tea consumption in group T (4.5 ± 0.9 vs 5.2 ± 0.9 , $p < 0.0001$). The CFVR ratio of group T was larger than that of group C (1.18 ± 0.07 vs 1.04 ± 0.08 , $p = 0.002$). Acute black tea consumption improves coronary vessel function, as determined by CFVR.